c. $f(x) = \sqrt{x-1}$ is continuous on $[1, \infty)$ since $\lim_{x \to 1^+} f(x) = f(1) = 0$

Check its continuity like left hand limit and right hand limit

For $f(x) = \sqrt{x-1}$, first note the **domain** is

$$x - 1 \geq 0 \;\Rightarrow\; x \geq 1,$$

so f is defined only on $[1, \infty)$.

1) At the endpoint $x = 1$

Right-hand limit

$$\lim_{x \to 1^+} \sqrt{x-1} = \sqrt{0} = 0.$$

Function value

$$f(1) = \sqrt{1-1} = 0.$$

So,

$$\lim_{x \to 1^+} f(x) = f(1),$$

which means f is **continuous at** $x = 1$ **(from the right)**, i.e. continuous on its domain at the endpoint.

Left-hand limit

$$\lim_{x \to 1^-} \sqrt{x-1}$$

does not exist as a real limit, because for $x < 1$, $x - 1 < 0\; explains$ and $\sqrt{x-1}$ is **not real** (and f is not defined there).

Conclusion at $x = 1$:

- LHL: does not exist (in $\mathbb{R}$)
- RHL: 0
- $f(1) = 0$
- Therefore, f is **right-continuous at 1**, and hence **continuous on** $[1, \infty)$ in the usual "continuity on a domain interval" sense (endpoint uses one-sided limit).

2) For any $x > 1$

$\sqrt{x-1}$ is a composition of continuous functions ($x - 1$ and $\sqrt{\cdot}$ on $(0, \infty)$), so it is **continuous for all** $x > 1$.

Final statement

$$f(x) = \sqrt{x-1} \text{ is continuous on } [1, \infty).$$

so i there is no interval then function is discontinuos

No. "No interval on the left" does **not** make it discontinuous.

Continuity is only checked **where the function is defined**.

- $f(x) = \sqrt{x-1}$ is defined only for $x \geq 1$.
- At the endpoint $x = 1$, we use the **right-hand limit** (because there are no domain points to the left).

So the correct endpoint test is:

$$\lim_{x \to 1^+} f(x) = f(1).$$

Here,

$$\lim_{x \to 1^+} \sqrt{x-1} = 0 \quad \text{and} \quad f(1) = 0,$$

so it **is continuous at** 1 (on its domain).

If someone insists on checking a **two-sided** limit at 1, then you'd say "the two-sided limit doesn't exist in $\mathbb{R}$ because the function isn't defined for $x < 1$." But that doesn't label it discontinuous—because continuity isn't required outside the domain.

Conclusion: $\sqrt{x-1}$ is continuous on $[1, \infty)$.